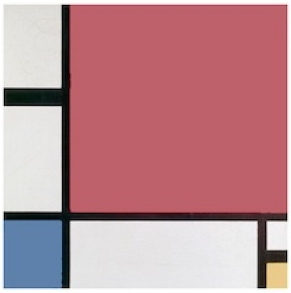
Visual Scene Graph

Baseline Experiment



Use a binding operation to encode the colored features in this image, and to bind those three features to the overall scene. A memory query either completes the scene (replacement update rule) or predicts the missing elements (difference update rule).

Requirements

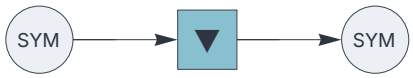
```
<< "Circuits.m"
```

Circuit


```
{
  "$schema": "https://creatingintelligence.org/schemas/circuit-v1.json",

  "hyperparameters": {"default": [2000, 10]},
```

```
  "dataflow": [
    {"component": "input", "plugin": "codec", "send": [11]},

    {"component": "auto", "send": [21], "receive": [11], "learn": true, "decimate": 0.95},

    {"component": "output", "plugin": "codec", "receive": [21]}
  ]
}
```

```
circ = Circuit[ json];
f = circ["function"];
circ["schematics"][ImageSize→ 220]
```



Train

```
f[ {"O1", "large", "red", "top", "right"}];
f[ {"O2", "medium", "blue", "bottom", "left"}];
f[ {"O3", "small", "yellow", "bottom", "right"}];
f[ {"Scene", "O1", "O2", "O3"}];
```

Test

```
f@ "Scene"
```

```
Sequence[{O1, O2, O3, Scene}]
```

```
f@ "O1"
```

```
Sequence[{large, O1, O2, O3, red, right, Scene, top}]
```

```
f@ "red"
```

```
Sequence[{large, O1, red, right, top}]
```

```
f@ "top"
```

```
Sequence[{large, O1, red, right, top}]
```

```
f@ "bottom"
```

```
Sequence[{blue, bottom, left, medium, O2, O3, right, small, yellow}]
```

```
f@ {"bottom", "right"}
```

```
Sequence[{bottom, O3, right, small, yellow}]
```

```
f@ {"blue", "medium"}
```

```
Sequence[{blue, bottom, left, medium, O2}]
```

```
f@ "left"
```

```
Sequence[{blue, bottom, left, medium, O2}]
```

```
f@ "right"
```

```
Sequence[{bottom, large, O1, O3, red, right, small, top, yellow}]
```

```
f@ "small"
```

```
Sequence[{bottom, O3, right, small, yellow}]
```

```
f@ {"Scene", "blue"}
```

```
Sequence[{blue, Scene}]
```